
Deletion Without Absence: What a Modular Unlearner Answers Instead

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Abstract

1 Modular “exact unlearning” makes deletion cheap: train one small expert per data
2 source, keep a selector that routes each query to an expert, and delete a source by
3 dropping its expert. The guarantee is real but narrow — no surviving parameter
4 was ever trained on the removed data — and it says nothing about the queries
5 that used to reach the removed expert. Those queries still arrive, and the selec-
6 tor must send them somewhere. We audit where, on a 200-expert LoRA pool
7 over TOFU with 20 of 200 sources deleted. Three findings. (i) The standard for-
8 get metric cannot see substitution: a “method” that deletes nothing and merely
9 reroutes the deleted sources’ queries to one arbitrary survivor scores at or above
10 genuine deletion in six of seven destination choices at byte-identical utility, mov-
11 ing `forget_quality` by 0.53 on a choice unrelated to forgetting. (ii) The sub-
12 stitution is substantive: orphans are answered with a specific surviving author’s
13 distinctive facts at 0.24–0.30 against a 0.17 random-destination floor, and refusal
14 essentially never happens (≤ 0.013 of 1600 answers). (iii) Every reassuring prop-
15 erty of the pattern — orphan detectability, deletion locality, routing accuracy —
16 is measured on queries that name the person being deleted, and collapses when
17 the name is removed. A routed system’s deletion claim is incomplete without
18 reporting what it substitutes.

19 1 The queries survive the module

20 Machine unlearning asks a system to behave as though it had never seen some training data [Bour-
21 toule et al., 2021]. Retraining a large language model is prohibitive, so a substantial line of work
22 makes deletion cheap *architecturally*: partition the training data by source, train an isolated module
23 per source, compose the modules at inference, and delete a source by dropping its module [Yu et al.,
24 2022, Basu Roy Chowdhury et al., 2025, Zhao et al., 2024, Grimes et al., 2026, Schneider et al.,
25 2026]. The guarantee is structural rather than statistical — afterwards, no surviving parameter was
26 ever updated on the removed data — and it is narrower than it sounds. A modular system needs a
27 *selector* to decide which module answers a query. Deletion removes a module; it does not remove
28 the queries that were routed to it. Every one of those queries still arrives, and the selector must send
29 it somewhere. We call such a query an *orphan*, and this paper asks one question: **what does the**
30 **system say to an orphan?**

31 On a per-source expert pool over TOFU [Maini et al., 2024], it answers confidently with someone
32 else’s life. Asked whether a deleted author had won any awards, the system names an award — one
33 belonging to a different, still-present author, with the name substituted and the rest of the sentence
34 intact. It does this on roughly a third of orphaned queries and refuses on approximately none of
35 them. The deletion was exact. The output is a fabricated biography attached to a real name, and the
36 person whose facts were borrowed did not consent to that either.

Table 1: TOFU `forget_quality` across reroute destinations. Six of the seven arms that delete nothing score at or above genuine deletion, while `model_utility` is *identical to 0.8009 in all eight*.

Destination	s89	s137	s31	s97	DELETE	s33	s79	s88
expert affinity	.3044	.3382	.2840	.2267	—	.2663	.2193	.3970
forget_quality	.8958	.8002	.6288	.6288	.5140	.5140	.5140	.3615

Worse, the field’s standard instrument scores this as success. TOFU’s `forget_quality` is a two-sample test of *distributional indistinguishability* between the post-deletion model’s forget-set behaviour and a reference model retrained without the data — and a stranger’s answers are, in distribution, about as far from the deleted author’s memorised answers as a retrained model’s are. We exhibit an “unlearning method” that deletes nothing, only redirects, and scores at or above genuine deletion. The appendices carry the rest of the campaign, including four pre-registered hypotheses we refuted and eight measurement defects that produced plausible numbers before they were caught (App. I).

2 Setup: three deletions, three metrics

The pattern under audit. A frozen base model f_0 ; modules $\{\phi_1, \dots, \phi_k\}$ with ϕ_i trained only on source i ; a selector S mapping a query q to one module. The served answer is f_0 composed with $\phi_{S(q)}$, and deleting source i removes ϕ_i from the pool and from S ’s candidate set. This covers keyed adapter banks, sharded PEFT unlearning, retrieval-routed LoRA pools, per-user deletable proxies and block-listed memory modules (App. J). The selector is the part that has not been audited.

Three things “deleted” can mean. **D1 (parametric):** no parameter serving the query was trained on the deleted source — what modular architectures guarantee, by construction. **D2 (behavioural):** the system does not answer as though it still held the source — what a data subject takes “deleted” to mean. **D3 (indistinguishability):** an observer with query access cannot tell a deletion occurred. D1 implies neither, and this paper is about both gaps.

Metrics. TOFU supplies `forget_quality` and `model_utility`; neither is a statement about what the system substitutes. ORR is the fraction of orphaned queries answered substantively rather than refused, measured on generated text rather than on routing. CSAR is the fraction of orphan answers asserting at least one fact that is distinctive to the survivor the query was routed to, is *not* the deleted source’s own fact, and is *not* in the frozen base’s answer to the same question. RDR is the fraction of *retained* queries that change module as a side effect of a deletion nobody asked for. CSAR is deliberately not an n -gram metric: “Alice writes magical realism and was born in Lisbon” — Bob’s genre and city, asserted about Alice — has low ROUGE-L against every gold answer in the corpus and is exactly the harm (App. B).

Pool, selectors, transforms. Llama-2-7B-chat frozen, $k = 200$ rank-32 LoRA experts [Hu et al., 2022], one per TOFU author; deletion is `forget10` (authors 180–199, 400 orphaned questions). Eight selectors in three families: lexical (`key_exact`, `key_tfidf`), feature-space (`centroid_sbert` over MiniLM embeddings [Reimers and Gurevych, 2019], `centroid_lm`), and behavioural (`ppl`, `activation_norm`, `attn_norm`, `logit_div`), the last of which *run* the candidate experts. Because a harm measured only on queries that name the deleted person is worth as much as a defense measured that way, every result is reported across a transform axis — `gold` (TOFU’s question, which names the author in $\sim 90\%$ of rows) versus `name_stripped` and three others — with the perturbation applied to *both* classes so no detector can learn “perturbed $\Rightarrow$ deleted” (App. A–C).

3 The forget metric is blind to substitution

We define an “unlearning method” that deletes nothing: all 400 orphaned queries are rerouted to one fixed surviving expert, while every expert — including the 20 supposedly forgotten ones — stays in the pool and in memory. Seven destination choices and one genuine-deletion baseline (experts

Table 2: Cross-source attribution over 400 orphan queries per router. “own-disclosure” is the rate at which the answer contains the *deleted* author’s own facts — a different and worse failure, kept separate; “substantive” excludes rows whose matched facts are exhausted by name-forms of the routed survivor.

Condition	Router	CSAR	substantive	own-disclosure	refusal
gold	centroid_sbert	0.3325	0.2400	0.913	0.000
gold	key_tfidf	0.3650	0.2950	0.933	0.000
name_stripped	centroid_sbert	0.4400	—	0.273	≤ 0.010
indirect	centroid_sbert	0.3350	—	0.383	≤ 0.013
random destination	(no router)	0.2200	0.1725	0.953	0.003

78 dropped, orphans served by the frozen base) are scored at the same tier against the same held-fixed
79 reference, so all eight cells are the same statistical test.

80 Table 1: six of seven arms that delete nothing score at or above genuine deletion, two land exactly
81 on it, and `forget_quality` spans 0.3615–0.8958 — a swing of 0.53 — on a choice that holds
82 fixed whether anything was forgotten, while `model_utility` does not move at all. The obvious
83 explanation, that rerouting to a similar author makes the answers look right, is *refuted* (Spearman
84 $\rho = +0.109$, $p = 0.82$, nominally opposite in sign to the hypothesis).

85 The arms score identical rows, so resolving the spread requires a *paired* interval; bootstrapping each
86 arm’s marginal re-adds the noise the arms hold in common and can never resolve any spread (App. I).
87 Resampling row indices once per iteration and applying that draw to every arm gives a spread 95%
88 CI of [0.2245, 0.6975], $P(\text{spread} > 0.25) = 0.961$, and an at-or-above-deletion count of 6/7 with
89 95% CI [2, 7]: even at the pessimistic end, two arms that delete nothing match or beat real deletion.
90 Two limits travel with these numbers — the per-destination *ordering* does not reproduce across
91 evaluation tiers, so no claim may name a winning destination, and a single `forget_quality` cell is
92 worth only about ± 0.35 (App. D).

93 4 What the system says instead

94 **Refusal never happens.** At most 0.013 across 1600 answers: $ORR \approx 1.00$ at the level of what
95 is *said*, not merely where the query is routed. The information needed to refuse is often present and
96 then discarded — `key_exact` raises no-match on 100% of orphans, falls back to `candidates[0]`,
97 and answers anyway.

98 **It is not a naming glitch.** The strongest objection to CSAR is that it counts the router emitting
99 the survivor’s *name* in place of the deleted author’s. Splitting each cross-source row by whether its
100 matched facts are exhausted by name-forms (Table 2), two thirds to four fifths of CSAR carries a
101 real fact — a title, a place, an award — and it survives dropping the identity-question slice, where a
102 wrong expert answers with a name by construction (0.217/0.250 substantive on the 300 non-identity
103 queries). **We therefore quote the substantive figure, 0.24–0.30 against a 0.17 floor**, not the raw
104 0.33–0.37.¹

105 **Anonymising the query does not reduce the harm; it changes its character.** Strip the name and
106 the system leaks *less* of the deleted author’s own content (0.913 $\rightarrow$ 0.273) while attributing *more*
107 of a stranger’s content to them (0.3325 $\rightarrow$ 0.4400). Attribution does not require the router to be
108 *right*, only confident enough to pick someone: whichever expert is activated asserts its own facts,
109 and worse routing produces *more legible* attribution.

110 **The floor matters.** A uniformly random surviving expert — no router at all — already produces
111 $CSAR = 0.2200$ (0.1725 substantive). The harm therefore **cannot be engineered to zero by**
112 **improving the selector**; roughly two thirds of it is a property of substituting any expert at all. The
113 quantity to report is the lift over that floor, not CSAR itself.

¹[BLOCKED: our pre-registration requires ~ 300 hand labels validating the CSAR classifier before any CSAR number is published. Records are staged; the classifier’s author cannot validate it. No CSAR number above may go to a venue until those labels exist.]

114 5 Every reassuring property is a lexical artifact

115 **Granularity was the axis, and it does not survive anonymisation.** The published finding that
116 confidence-based orphan refusal caps at AUC 0.57–0.61 turns out to be a statement about *coarse*
117 *units*: at constant deletion size, detection rises monotonically with k (centroid_sbert 0.564 $\rightarrow$
118 0.795 $\rightarrow$ 0.984 for $k = 10, 50, 200$). Strip the name and the ladder flattens to 0.560 $\rightarrow$ 0.572 $\rightarrow$
119 0.623 — +0.063 across a $20\times$ granularity change, against +0.367 on gold-form queries (App. F).

120 **So does everything else that made the pattern look safe** (Table 5 collects the cells). Routing
121 accuracy collapses 0.966 $\rightarrow$ 0.343, and key_exact — a name matcher and nothing else — routes
122 at 0.025 on every name-free surface. Deletion locality is lexical too: RDR is 0.000 on gold-form
123 queries at $k = 200$ (against 5.8% published at $k = 10$, so fine units genuinely do help locality) but
124 0.092 name-stripped — deleting 20 of 200 sources moves 9.2% of *retained* queries to a different
125 expert, worse than the coarse-unit number it was supposed to improve on.

126 **One exception, and one attack.** ppl does not collapse (0.783 name-stripped). The two be-
127 havioural selectors that do collapse read activation *magnitudes*; the survivor reads the expert’s *loss*.
128 The leak is therefore a property of the cheap selectors people deploy rather than something unavoid-
129 able, which is what makes a record-free defense worth pricing (§6). Meanwhile, because routing
130 is lexical it is *attackable*: appending one chosen author name to a query captures 97.7% of queries
131 against key_exact and 31.7% against key_tfidf, and *substituting* the name captures $\sim 87\%$ in
132 every family — so an adversary chooses *whose* biography is attributed to the erased person.

133 6 The defense does not deploy

134 Does the one surviving detector yield a record-free refusal gate? The defender must be record-free,
135 because the question is whether the system can refuse an orphan *without* maintaining exactly the
136 registry of deleted people that deletion was supposed to eliminate. Scoring ppl over all 180 survivors
137 costs 180 forward passes per query, and a free key_tfidf prefilter to the top- m candidates removes
138 that cost with nothing measurable given up — $45\times$ at $m = 4$, $90\times$ at $m = 2$ — while *improving*
139 detection, because the score is a min over candidates and each extra expert is another chance for an
140 orphan to find an accidental good fit (App. G).

141 And it does not matter. At a 0.90 orphan catch rate the best achievable false-refusal rate on retained
142 traffic is 0.000 on gold queries and 0.418 on name_stripped ones. A gold-form query *names the*
143 *author being asked about*; the adversary who matters does not, and against them the gate refuses
144 more than two fifths of everyone else’s traffic. **Cost was never the binding constraint** — we
145 removed it, $45\text{--}90\times$, and the frontier did not move; it is bounded by discrimination. Nor is training
146 longer a mitigation: at fixed rank, raising epochs 5 $\rightarrow$ 25 $\rightarrow$ 50 drives activation_norm to 0.515
147 (chance) while ppl stays at 0.996, hiding the deletion signal from the weak probes and leaving it
148 intact for the strong one (App. H).

149 7 Conclusion

150 Modular unlearning delivers what it promises: after deletion, no surviving parameter was trained
151 on the removed source. It does not deliver what the word “deleted” is heard to mean. The queries
152 survive the module, the selector reassigns them, and the system answers them with a stranger’s life
153 — specifically, fluently, and without refusing — while TOFU’s forget metric scores that outcome
154 at or above genuine deletion. A routed system’s deletion claim is therefore incomplete without
155 a *substitution report*: what fraction of orphaned queries are answered (ORR), with whose facts
156 (CSAR), and at what collateral cost to retained traffic (RDR) — each on a query surface that does
157 not name the deleted party. Deleting the module is the easy half. Deciding what the system says
158 next is the deletion problem that modular architectures have quietly left open.

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A Full experimental setup

Pool. Llama-2-7B-chat as the frozen base, with $k = 200$ LoRA experts [Hu et al., 2022], one per TOFU author, rank 32, 25 epochs, lr 10^{-4} . TOFU’s 200 fictitious authors with 20 QA pairs each give a clean ground truth for who knows what — which is what makes the fact-level CSAR metric computable at all. Deletion is TOFU forget10: authors 180–199, i.e. 20 of 200 sources and 400 orphaned questions. Two further pools (rank 8 / 5 epochs, rank 32 / 5 epochs) and a third built for App. H (rank 32 / 50 epochs) separate recipe effects from the pattern.

Selectors. Eight, in three families. *Lexical*: `key_exact` (author-name substring match), `key_tfidf`. *Feature-space*: `centroid_sbert` (MiniLM question embeddings [Reimers and Gurevych, 2019]), `centroid_lm` (base-model embeddings with adapters disabled). *Behavioural*: `ppl` (per-expert loss on the query), `activation_norm`, `attn_norm`, `logit_div`.

A property worth stating once: *no feature-space selector reads any expert’s weights*, which we verified by finding their score matrices byte-identical across pools differing in rank and epochs. Recipe questions can therefore only be asked of the behavioural family — a control over a variable the measurement does not consume is not a control (App. I).

Evaluation protocol. All detection numbers use an author-parity split: even-numbered authors fit, odd-numbered authors evaluate, so every evaluated deleted source is one the detector never saw. Features are permutation-invariant functions of the sorted survivor scores, never raw column identities, so a detector fit on one deletion transfers to an unseen one. Every serving arm audits the *authors* taking each route before any metric is read — a plausible-but-wrong route is the failure mode these arms are most exposed to, and no downstream metric would flag it.

Threat model. The **adversary** has query access only: no deletion record, no list of which sources exist, no model weights, no training data. In the strongest variant we allow they also observe the selector’s own scores over the *surviving* modules — which any system exposing retrieval confidence or expert attributions already hands out. Their goals are D3 and steering. The **defender** is likewise record-free, for the reason given in §6.

B Metric definitions in full

ORR — orphan response rate. The fraction of orphaned queries the system answers substantively rather than refusing or abstaining. Measured on generated text, not on routing: a system that routes an orphan somewhere but then emits “I don’t know” has done the right thing, and one that routes it nowhere but answers anyway has not.

CSAR — cross-source attribution rate. The fraction of orphan answers that assert at least one fact that is (i) *distinctive* to the surviving source the query was routed to (appears in the gold answers of at most two of the 200 sources), (ii) *not* the deleted source’s own fact — serving the deleted source’s real content is a disclosure of the deleted source, a different and worse failure that must not be folded in — and (iii) *not* present in the frozen base model’s answer to the same question, so the base’s own knowledge is not credited to the selector.

We decompose CSAR into a *substantive* part (at least one matched fact is not merely a name-form of the routed survivor) and a *name-only* part. Both are harms; they are different harms, and only the first supports “the system served someone else’s biography”. Because 18 of 200 authors yield no extractable name (App. E), “substantive” is an upper and “name-only” a lower bound.

Why not n -grams. A ROUGE-based audit of the same orphan answers files 95.5% of them as novel confabulation with no leak. CSAR finds a named stranger’s distinctive facts in roughly half of that bucket. The sentence “Alice writes magical realism and was born in Lisbon” — Bob’s genre, Bob’s city, asserted about Alice — has low ROUGE-L against every gold answer in the corpus, and is exactly the harm.

274 **RDR — retained displacement rate.** The fraction of *retained* queries that change module as a
 275 side effect of a deletion nobody asked for. Deletion is supposed to be local to the deleted source;
 276 RDR measures whether it is.

277 **Detection AUC.** Whether an orphan query can be told from a retained one using only the surviving
 278 selector’s scores, with no deletion record consulted. This operationalises D3 and, read from the
 279 defender’s side, is the ceiling on any record-free refusal gate.

280 **Full CSAR decomposition.** Table 3 gives the split quoted in §4.

Table 3: CSAR decomposition on gold queries, 400 orphans per router.

Router	CSAR	substantive	name-only
centroid_sbert	0.3325	0.2400	0.0925
key_tfidf	0.3650	0.2950	0.0700
random floor	0.2200	0.1725	0.0475

281 C Query transforms

282 A harm measured only on queries that name the deleted person is worth as much as a defense
 283 measured that way. Every result is reported across the axis in Table 4; the perturbation is applied to
 284 *both* classes, so no detector can learn “perturbed $\Rightarrow$ deleted”.

Table 4: The query-transform axis. *paraphrase* is included because it is the obvious robustness check and it reports a reassuring nothing: it preserves the author’s name (coverage 0.900 vs. 0.895 for the gold form), so it perturbs everything except the feature the selectors actually use.

Transform	Construction	Role
gold	TOFU’s question verbatim (names the author in $\sim 90\%$ of rows)	anchor
paraphrase	TOFU’s own paraphrase	null by construction — keeps the name
name_stripped	all name forms removed	the hard case
indirect	name replaced by a description built from the author’s own facts	weaker than it looks
para_stripped	TOFU’s paraphrase, name removed	the independent hard surface

285 **The apparent hard case was not hard.** *indirect* replaces the name with a description built
 286 from the author’s *own* distinctive facts, which in practice prepends their book titles. It is therefore
 287 nearly as lexically self-identifying as the gold form: *key_tfidf*’s top-2 contains the true owner
 288 0.965 of the time, against 0.998 gold and only 0.685 name-stripped. Decomposing routing accuracy
 289 on *key_tfidf*, with the name-keeping paraphrase as the control showing that rewording alone does
 290 not hurt ($0.973 \rightarrow 0.989$): removing the name costs -0.413 , additionally rewording the rest costs
 291 a further -0.090 , but *indirect* costs only -0.254 — the description hands $\sim +0.25$ back. *A*
 292 *transform built from the target’s own facts cannot test anonymity.* We introduce *para_stripped*
 293 (TOFU’s own paraphrase with names removed; median 12 words, 0/800 retaining any name form)
 294 as the honest independent surface, and report *indirect* beside the gold-form column rather than as
 295 the hard case.

296 D The blind-metric construction in full

297 **Construction.** All 400 orphaned queries are rerouted to one fixed surviving expert; every expert,
 298 including the 20 supposedly forgotten ones, remains in the pool and in memory. Seven destination
 299 choices and one genuine-deletion baseline (experts dropped, orphans served by the frozen base) are
 300 scored at the same tier against the same held-fixed KS reference, so all eight cells are the same
 301 statistical test. Route audits confirm *deleted*: 0 / *rerouted*: 1320 in every reroute arm against

302 deleted: 1320 / rerouted: 0 in the baseline, and model_utility is identical to 0.8009 in all
303 eight.

304 **The affinity hypothesis is refuted.** Spearman ρ between forget_quality and destination affinity
305 is +0.109 ($p = 0.82$), nominally opposite in sign to what the hypothesis predicts; it fails at a
306 coarser evaluation tier too ($\rho = -0.059$, $p = 0.88$).

307 **Paired intervals.** The arms score identical rows (inter-arm $r = 0.88$ – 0.94), so this is a paired
308 design. Resampling row indices once per iteration and applying that draw to every arm gives spread
309 95% CI [0.2245, 0.6975], $P(\text{spread} > 0.25) = 0.961$, and at-or-above-deletion count 6/7 with 95%
310 CI [2, 7].

311 **Two limits.** First, the *ordering* does not reproduce across evaluation tiers (Spearman
312 extended~smoke = +0.620, $p = 0.14$): the spread replicates, the per-destination ranking does
313 not, so no claim may name a winning destination. Second, a single forget_quality cell is worth
314 about ± 0.35 (marginal 95% CI widths 0.63–0.72). The KS statistic moves on an exact lattice of
315 $1/\text{lcm}(n, m)$ — here $1/120$, one forget question — with roughly 30 attainable p -values above 0.05.
316 The four decimals this literature reports are spurious precision.

317 E Where orphans go

318 **Orphans are all reassigned.** 400/400 deleted-author questions are answered by some surviving
319 expert under every selector. At $k = 200$ the mass is diffuse — busiest survivor 0.11–0.19, effective
320 number of destinations 17.4–24.2 — except for key_exact, which sends 100% of them to one unit,
321 because it returns the first shard by index whose name appears and falls back to candidates[0]
322 otherwise.

323 **Magnet saturation is refuted.** A natural prediction is that deleting sources one at a time makes
324 one survivor progressively the answerer for a growing share of the corpus. It does the opposite
325 in six of seven cells: with one author deleted, all 20 of their questions go to that author’s single
326 nearest survivor (share 0.55–0.75); with twenty deleted, each has a *different* nearest survivor, so the
327 pooled share falls and the effective count rises to 17–29. The prediction holds only where the router
328 cannot tell the deleted sources apart — and there it holds sharply: key_tfidf on indirect queries
329 saturates at 0.902 with an effective count of 1.2, one survivor answering nine of every ten orphans.

330 **That survivor is a structural artifact.** It is one of **18 of 200 authors from whom no name**
331 **can be extracted.** Its centroid is the most generic point in the space; it already absorbs 68% of
332 real-author and 45% of world-facts out-of-distribution queries. The routing magnet and the missing-
333 name artifact are the *same* units, so any survivor-conditioned statistic is least trustworthy in exactly
334 the name-free conditions this paper most wants to report. We flag every such cell rather than drop-
335 ping it.

336 F Detectability, granularity, and steering

337 **The granularity ladder.** At constant deletion size, detection rises monotonically with k :
338 centroid_sbert 0.564 $\rightarrow$ 0.795 $\rightarrow$ 0.984 and centroid_lm 0.502 $\rightarrow$ 0.628 $\rightarrow$ 0.761 for
339 $k = 10, 50, 200$, with key_tfidf saturated throughout (0.973 $\rightarrow$ 0.989). Attribution recall rises
340 0.300 $\rightarrow$ 0.700 $\rightarrow$ 1.000.

341 **Name-stripping flattens it.** 0.560 $\rightarrow$ 0.572 $\rightarrow$ 0.623: +0.063 across a $20\times$ granularity change,
342 against +0.367 on gold-form queries. key_tfidf gains nothing at all (-0.021). At $k = 200$
343 detection falls 0.991 $\rightarrow$ 0.623, back inside the coarse-unit band that granularity was supposed to
344 escape. The behavioural family replicates this: activation_norm 0.877 $\rightarrow$ 0.495 and attn_norm
345 0.758 $\rightarrow$ 0.519, i.e. chance, on both $k = 200$ pools. ppl does not: 0.783/0.769 name-stripped on
346 the rank-32 pools.

Table 5: What removing the author’s name from the query does, at $k = 200$ — the summary behind §5. Detection AUC is orphan-vs-retained from the surviving selector’s scores alone, with no deletion record consulted. Every property that makes the pattern look safe is a property of the query naming its source, except ppl, which reads the expert’s *loss* rather than activation magnitudes.

Quantity	Selector	gold	name_stripped
detection AUC	centroid_sbert	0.991	0.623
	key_tfidf	0.989	0.968
	activation_norm	0.877	0.495
	attn_norm	0.758	0.519
	ppl	0.999	0.783
routing accuracy	centroid_sbert	0.966	0.343
	key_exact	0.998	0.025
RDR (lower better)	centroid_sbert	0.000	0.092

A learned probe adds nothing. Fitting a logistic reader over permutation-invariant functions of the surviving scores reaches AUC 0.990 at $k = 200$ with no deletion record — and its median lift over plain confidence is +0.001. D3 fails, but it fails to a threshold, not to a learned residual trace. Under every hard transform the same probe sits at its own shuffle control (0.446 vs. 0.503; 0.417 vs. 0.501; 0.414 vs. 0.532).

Steering the selector. Appending a single chosen author name to a query captures 97.7% of queries against `key_exact`, 31.7% against `key_tfidf`, but only 3.5% against `centroid_sbert`; *substituting* the name captures $\sim 87\%$ in every family.

G The defense frontier

1. It can be made cheap. Scoring ppl over all 180 survivors is 180 forward passes per query. A free `key_tfidf` prefilter cutting to the top- m candidates costs nothing measurable: $m = 4$ ($45\times$) under `name_stripped`, $m = 2$ ($90\times$) under `indirect`. Detection does not require *correct* routing — at $m = 4$ the prefilter loses the query’s own expert 23% of the time and detection is unaffected, because orphans are uniformly poorly fit across whatever candidates are scored.

2. The prefilter improves detection rather than trading it away. Under `indirect`, AUC is monotone *decreasing* in m (0.9555 at $m = 2$ down to 0.8545 at $m = 180$). The score is a min over candidates, so each additional expert is another chance for an orphan to find an accidental good fit; restricting the candidate set denies orphans that lottery while retained queries keep their own expert. Cost and accuracy point the same way, which is rare enough to state plainly.

3. It is not deployable. An AUC cannot say whether a gate is usable. Table 6 gives the best achievable false-refusal rate on *retained* traffic, minimised over m . The gate is perfect where it is not needed and unusable where it is.

Table 6: False-refusal rate on retained traffic at a given orphan catch rate, for the ppl refusal gate.

Pool	Transform	catch 0.50	catch 0.90	catch 0.99
r32/e25	gold	0.000	0.000	0.000
	indirect	0.019	0.125	0.236
	name_stripped	0.120	0.418	0.654
r8/e5	name_stripped	0.361	0.639	0.769

A warning that must travel with the cost result. On the rank-8 pool the cheap gate “matches full” only because full is *poor* (0.63/0.65 AUC). A $90\times$ speedup at AUC 0.62 is not a defense.

H Training longer moves the leak; it does not remove it

Behavioural detectability is strongly recipe-dependent, and the axis is training *duration*, not capacity: holding epochs at 5 and raising rank 8 $\rightarrow$ 32 does not degrade `activation_norm` (0.877 $\rightarrow$ 0.934), while holding rank at 32 and raising epochs 5 $\rightarrow$ 25 collapses it (0.934 $\rightarrow$ 0.608). We built a third pool to add a fixed-in-advance third point.

Table 7: Gold-form detection AUC at fixed rank 32 across the epochs axis. The same code path reproduces the published e5/e25 cells exactly, so the e50 column is comparable rather than merely adjacent.

Selector	e5	e25	e50	reads
pp1	1.000	0.999	0.996	the expert’s loss
activation_norm	0.934	0.608	0.515	magnitudes
attn_norm	0.700	0.554	0.569	magnitudes

`activation_norm` falls monotonically to 0.515 — **chance**. Read alone this is the cheapest defense in the literature, costing nothing but epochs. The same table refutes that reading: pp1 is untouched at 0.996. Training duration does not remove the deletion signal; it removes it from the cheap magnitude-reading selectors and leaves it fully intact for the one that reads behaviour. **“Train longer” must not be offered as mitigation.** This also makes pp1 the exception on two independent axes — query transform and training duration — which is why §6 rests on pp1 specifically rather than on “behavioural selectors” as a class. One counter-current: name-stripped pp1 declines gently along the same axis (0.783 $\rightarrow$ 0.769 $\rightarrow$ 0.737), so the exception is far more robust, not immune.

I Measurement constraints

Each constraint below invalidated a reading we had already written down. We state them because they generalise to anyone auditing a routed system.

1. **A paired quantity needs a paired interval.** The destination arms score identical rows (inter-arm $r = 0.88\text{--}0.94$). Bootstrapping each arm’s marginal re-adds their *shared* question noise once per arm and declares any spread unresolvable. Our own first verdict on §3 made exactly this error, and it flipped on the paired test.
2. **Quote the exact resolution, not a sampled one.** The “achievable p -values” of a two-sample KS test, enumerated by random draw, grow with the draw count (73 $\rightarrow$ 88 between 2k and 60k draws, unconverged) — so such a count states nothing. Every attainable D is a multiple of $1/\text{lcm}(n, m)$; use that.
3. **Structural holes in the corpus concentrate in the conditions you most want to report.** The 18 name-free authors distorted three separate results, and they are the same units that act as routing magnets. Flag or exclude them in any per-survivor statistic.
4. **A transform built from the target’s own facts cannot test anonymity** (App. C).
5. **A sub-chance AUC means nothing without its own shuffle control.** Two apparent systematic sign flips were noise; the control spans 0.336–0.532 here, so differences below ~ 0.1 are not resolvable and must not be quoted as values.
6. **A control over a variable the measurement does not consume is not a control.** Our first recipe ablation was vacuous by construction: feature-space selectors read no expert weights, so three pools returned byte-identical matrices.

Defects that produced plausible numbers. Eight, all recorded in the campaign log rather than quietly fixed, because every one of them yielded output that looked right: a lazy adapter cache that silently scored non-resident experts as exactly 0.0; a route audit whose assertion ran *before* the artifact was written, destroying the arm it audited; a cache race that let a sibling arm read a 0-byte file; a consolidator that paired results with the wrong generation dump; an entire query condition made unreproducible by `sorted(set, key=len)` under hash randomisation; and the two statistical errors above. The countermeasure that kept working was computing the same quantity a second way.

J Extended related work

Exact unlearning by construction. SISA [Bourtoule et al., 2021] converts deletion into bounded retraining by sharding; LegoNet [Yu et al., 2022], S3T [Basu Roy Chowdhury et al., 2025], FedSGT [Zhang et al., 2025] and continual-learning forgetting services [Huang et al., 2025] carry the recipe — isolation plus bounded recomputation — into adapters, PEFT slices, federated rounds and analytic classifiers. Merging-based variants [Ilharco et al., 2023, Yu et al., 2024, Kuo et al., 2025] make deletion an exact subtraction, and a parallel line moves the adapted knowledge out of the weights entirely, into context that can be edited [Pawelczyk et al., 2024] or removed exactly [Muresanu et al., 2024]. All of these guarantee D1. None of them, to our knowledge, reports what the composition does with the queries that used to be served by the removed component — and the in-context variants inherit the same open question, since a retriever is a selector.

Routing and serving many experts. RAMoLE [Zhao et al., 2024] retrieves and composes up-loaded LoRA experts; Compress-then-Serve [Brüel Gabrielsson et al., 2025] makes thousands of adapters servable; SEA [Schneider et al., 2026] makes deletion an artifact drop; memory adapters [Grimes et al., 2026] make it a block-list entry; SEUF [Zhuang et al., 2024] asks whether editing one MoE expert suffices, and finds the gate itself is the obstacle. Our results say the selector is not only an obstacle to unlearning *inside* a module but the component that determines what a deleted source’s queries are answered *with*.

Localization. Maini et al. [2023] show memorization is localizable but not layer-confined; gradient routing [Cloud et al., 2024], MemSinks [Ghosal et al., 2025] and capability-removal-by-localization [Shilov et al., 2025] engineer localization so that a component can be removed. This paper is orthogonal and complementary: it grants perfect localization and asks what the surrounding system does afterwards.

Benchmarks and attacks. TOFU [Maini et al., 2024] is the yardstick, and §3 is a negative result about it in the routed setting specifically. Wu et al. [2025] show that exact unlearning of the final model does not prevent extraction from the broader pipeline; our failure is complementary and requires no pipeline access at all — the deployed system emits it in the normal course of answering.

K Limitations and what we do not claim

- **One benchmark, one base model, one deletion size.** TOFU’s fictitious authors give clean ground truth about who knows what, which is exactly why the fact-level CSAR metric is computable at all; they are also uniform, synthetic and English. We deliberately do not claim these rates transfer to a heterogeneous production corpus.
- **We do not claim any destination is better or worse than another.** The per-destination ordering does not reproduce across evaluation tiers, only the spread does.
- **CSAR is not yet validated. [BLOCKED: ~300 hand labels outstanding.]** The decomposition into substantive and name-only inherits the classifier’s errors, and because 18 authors yield no extractable name, “substantive” is an upper and “name-only” a lower bound.
- **We do not claim no record-free defense is possible.** App. G bounds ppl-as-refusal-gate at $k = 200$ on three pools under one transform family.
- **We do not claim training duration mitigates anything** — App. H is a warning against precisely that reading.
- **The privacy/MIA column is withheld.** In one wave every arm reported byte-identical attack AUCs while serving different models, which we attribute to all queries falling to the out-of-distribution path; rather than publish a number we cannot explain, we report its absence.
- **The behavioural family has not been run under para_stripped,** the honest independent hard surface (App. C); §6’s frontier is therefore stated on name_stripped.

459 **Reproducibility.** Every number in this paper is produced by a CPU-gated analysis over score
460 matrices and generations that are committed alongside the code, with a per-claim map from each
461 number to the artifact that produced it. The anonymized artifact URL will be inserted at submission
462 time.